
BIOLOGY

9648/01

Paper 1 Multiple Choice

22 September 2014
Monday

1 hour 15 mins

Additional Materials: Multiple Choice Answer Paper

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, PDG and identification number on the Answer Sheet.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

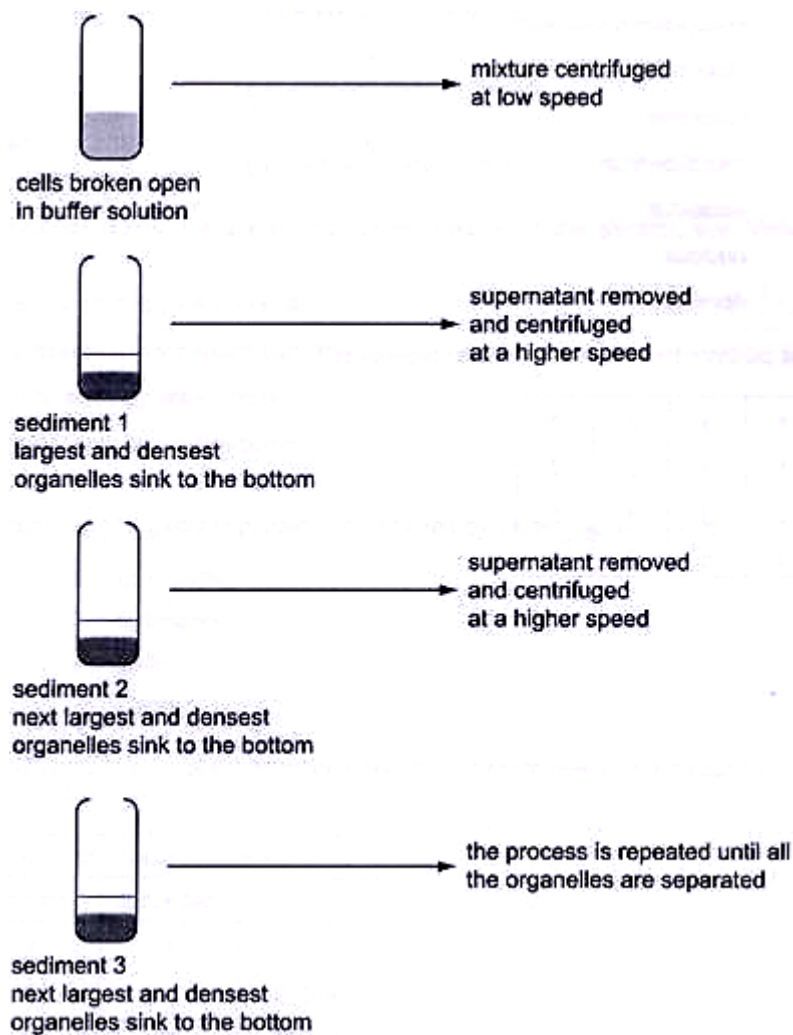
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

- 1 Fractionation is a process used to separate cell components according to their size and density. The diagram shows the main stages in fractionation of a plant cell.

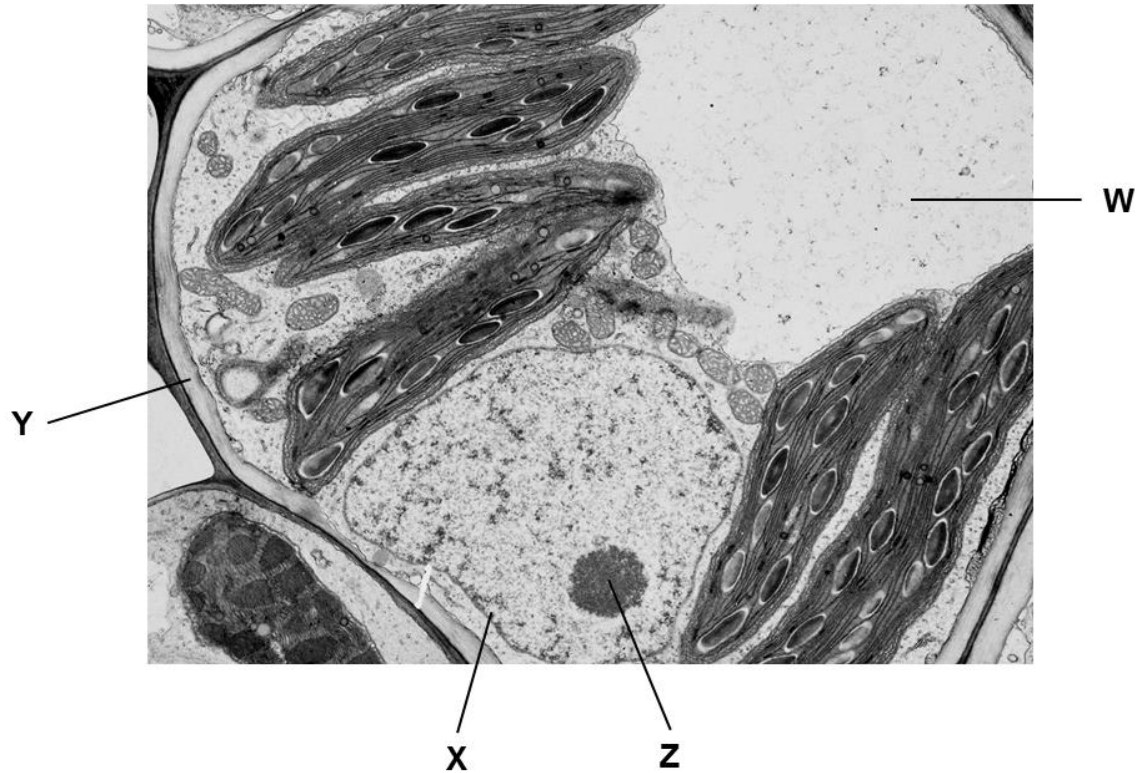


DCPIP and buffer solution (containing glucose, fructose, sodium bicarbonate) were added to each of the sediments, and the mixtures were left in the dark for fifteen minutes. Sediment 2 caused the DCPIP to be reduced.

Which organelle present in Sediment 2 caused reduction of DCPIP?

- A chloroplast
- B mitochondria
- C nuclei
- D ribosomes

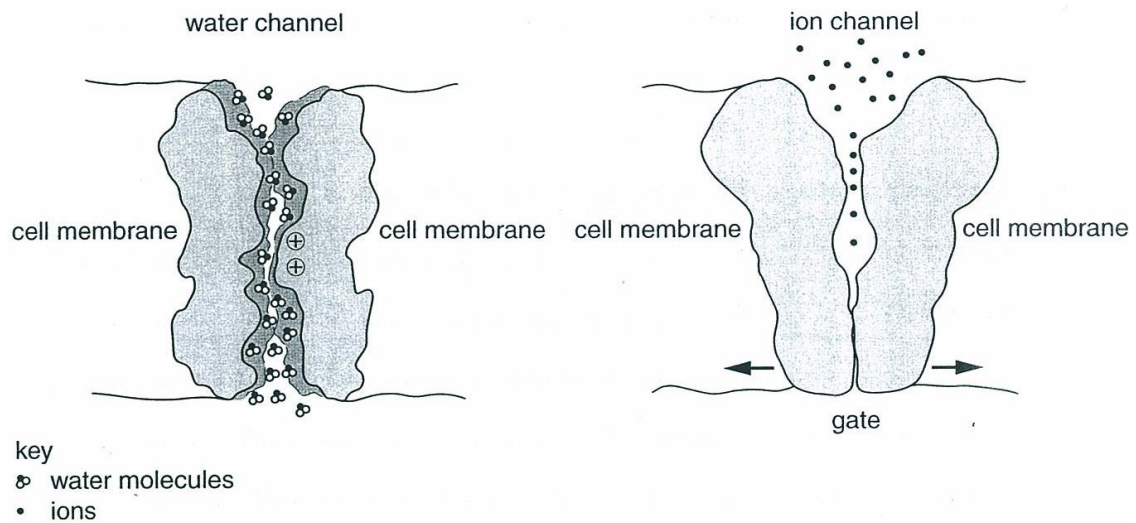
- 2 The electron micrograph shows a plant cell with structures W to Z labelled.



Which of the following structure to function relationships is correct?

	Structure	Function
A	W	Synthesis of carbohydrates
B	X	Gives the cell a fixed shape
C	Y	Regulates the movement of substances
D	Z	Synthesis of rRNA and assembly of ribosomal subunits

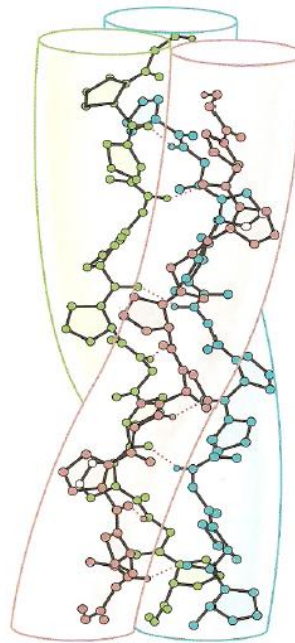
- 3 The figure shows water and ion channels that are found in the cell surface membrane of all cells.



Which of the following statements is true?

- A Movement of water molecules through the water channel requires energy provided by the hydrolysis of ATP as water molecules are polar while the phospholipid bilayer of the membrane is hydrophobic.
- B Common amino acid residues found on the protein surface surrounding the pores of both channels include valine and phenylalanine.
- C Only the ion channel allows for the regulation of ion movement across the cell surface membrane.
- D The ion channel is an example of a carrier protein as it is able to switch between two different conformations to allow the movement of ions.

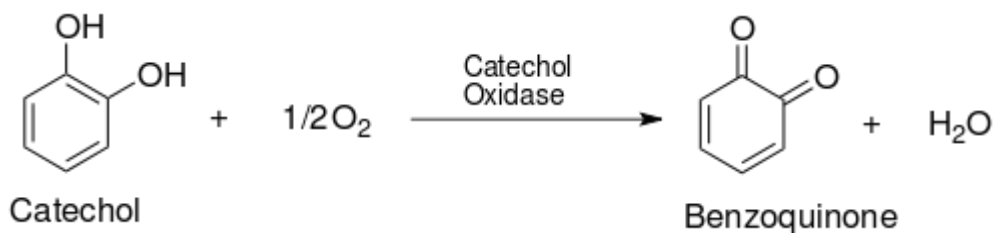
- 4 The molecule shown below is found in various parts of the human body, and is helical in structure.



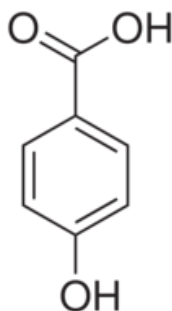
What else is true of this molecule?

- A** Hydroxyl groups projecting in all directions from each chain allow hydrogen bonds to form and result in bundling of the chains to form microfibrils.
- B** Double bonds present in each chain allow the chains to adopt helical structure.
- C** Complementary base pairing occurs through the formation of hydrogen bonds between chains.
- D** The small size of the R-groups of amino acid residues in each chain allows the chains to come close together to form hydrogen bonds.

- 5 Catechol is oxidised to benzoquinone, as shown in the equation, resulting in darkening of peeled fruits.



Catechol oxidase is an enzyme which is inhibited by parahydroxybenzoic acid (PHBA). The structure of PHBA is shown below.

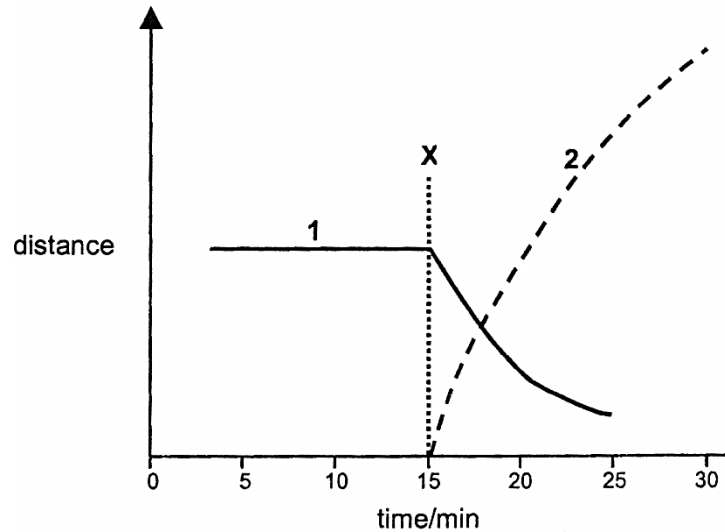


Catechol oxidase is also inhibited by phenylthiourea (PTU) which binds to a copper atom in the enzyme. The copper atom is essential for the oxidative activity.

Which of the following statements are **not** correct?

- I PHBA acts as a competitive inhibitor because its structure is similar to benzoquinone.
 - II PHBA acts as a competitive inhibitor, and in the presence of PHBA, the Michaelis constant, K_M of the reaction can be restored by increasing the concentration of catechol.
 - III PTU acts as a non-competitive inhibitor because it does not change the shape of the active site of catechol oxidase.
 - IV PTU acts as a non-competitive inhibitor, preventing the formation of enzyme-substrate complex between catechol oxidase, catechol and O_2 .
- A I and II only
- B III and IV only
- C I, II and III only
- D I, II, III and IV

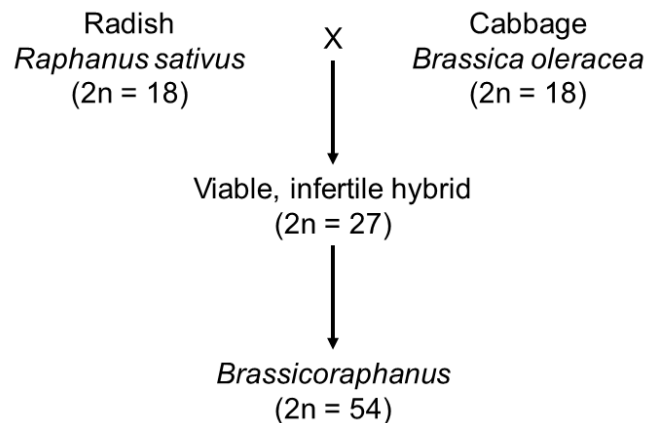
- 6 The graph below shows measurements taken either during meiosis I or meiosis II.



Identify the stage of meiosis that commences at X and the measurements that are illustrated by Graphs 1 and 2.

	Stage of Meiosis at X	Graph 1	Graph 2
A	Anaphase I	Distance between centromeres of homologous chromosomes and centrosome	Distance between centromeres of sister chromatids
B	Anaphase I	Distance between centromeres of chromosomes and centrosome	Distance between centromeres of homologous chromosomes
C	Anaphase II	Distance between centromeres of chromosomes and centrosome	Distance between centromeres of sister chromatids
D	Anaphase II	Distance between centromeres of homologous chromosomes and centrosome	Distance between centromeres of homologous chromosomes

- 7 A cross between radish (*Raphanus sativus*) and cabbage (*Brassica oleracea*) produced the following results.



Which of the following statements are true?

1. Non-disjunction of all sister chromatids during anaphase II resulted in the production of diploid gametes in radish, which fused with a haploid gamete of cabbage to give rise to the infertile hybrid progeny.
2. The hybrid is infertile because it has a diploid number which is odd.
3. The hybrid is infertile as there are nine unpaired chromosomes during prophase I, which prevented the formation of gametes.
4. Addition of colchicine prevents the formation of spindle fibres, which enabled the infertile hybrid to form gametes. Fusion of two of these gametes gives rise to *Brassicoraphanus*.

- A** 1 and 4
- B** 2 and 3
- C** 1, 2 and 4
- D** 1, 3 and 4

- 8 Which of the following statements about the newly synthesized strand of a human chromatid is correct?

- A** It was synthesised from multiple origins solely on continuous DNA synthesis.
- B** It was synthesised from multiple origins solely on discontinuous DNA synthesis.
- C** It was synthesised from multiple origins by a mixture of continuous and discontinuous DNA synthesis.
- D** It was synthesised from multiple origins by either continuous or discontinuous DNA synthesis, depending on which specific sister chromatid is being examined.

9 The descriptions below are of nucleic acids in eukaryotes.

1. A polynucleotide of variable length formed by base pairing.
2. A small polynucleotide with a specific three-dimensional shape.
3. A large polynucleotide with a specific shape associated with proteins.
4. A large polynucleotide with super coiled sections associated with proteins.

Which row correctly matches each description to its function?

	Stores coded information	Carries coded information	Carries specific amino acids	Provides a site for protein synthesis
A	3	4	1	2
B	3	4	2	1
C	4	1	2	3
D	4	3	1	2

10 Which of the following shows the possible effects of a single nucleotide substitution in each of the following locations in a gene on the production of the protein product of that gene?

	Promoter	Start codon	Stop codon	Middle of an intron
A	No protein product is produced	Protein product is longer than normal	Protein product is normal	Too much protein product is produced
B	Too much protein product is produced	No protein product is produced	Protein product is longer than normal	Protein product is normal
C	Protein product is normal	Protein product is shorter than normal	Too much protein product is produced	Protein product is longer than normal
D	Protein product is longer than normal	Protein product is normal	Protein product is shorter than normal	No protein product is produced

11 How do viruses cause disease in animals?

1. They inhibit normal host cell DNA, RNA, or protein synthesis.
2. They disrupt and inactivate the tumor suppressor genes of the host cell causing uncontrolled cell division.
3. They disrupt and inactivate the oncogenes of the host cell causing uncontrolled cell division.
4. Their viral proteins and glycoproteins on the surface membrane of host cells cause them to be recognised and destroyed by the body's immune defenses.
5. They deplete the host cell of cellular materials essential for metabolic functions.

- A** 1 and 5
B 2, 3 and 4
C 1, 2, 4 and 5
D All the above

12 A new antiviral drug has been developed to treat influenza A patients. It works by preventing progeny virions from detaching from infected cells. Which of the following components of the influenza virus does the new drug act on?

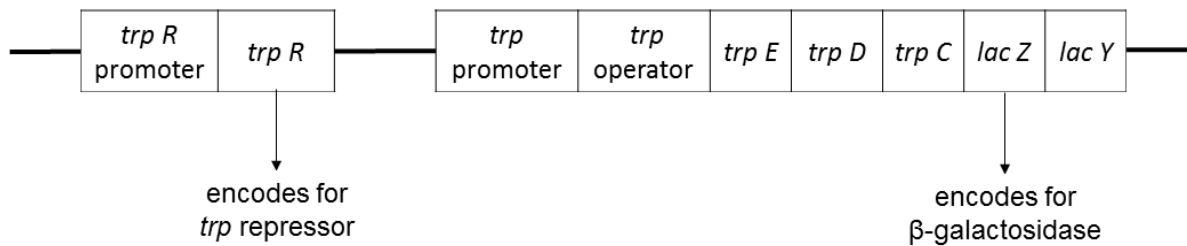
- A** Hemagglutinin
B Matrix protein
C Neuraminidase
D Nucleoproteins

13 When a mutant strain of *Escherichia coli* was grown in liquid medium containing glucose and lactose, it was observed that both lactose and glucose concentrations decreased rapidly over time.

Which of the following can account for the observations?

- A** Change in cAMP binding site of catabolite activator protein, such that it can no longer bind to cAMP.
B Change in DNA binding domain of catabolite activator protein, such that it has increased affinity for the CAP binding site on the promoter of the *lac* operon.
C Change in the allosteric site of the *lac* repressor, such that it can no longer bind to lactose.
D Mutation in the operator, such that *lac* repressor is always bound to it.

- 14 The diagram shows a *trp-lac* fusion operon, which is constructed by fusing of part of the *lac* operon to the *trp* operon.



Which statement correctly describes the effect of this fusion on β -galactosidase expression in bacteria cells?

- A β -galactosidase expressed is non-functional as the polypeptide is joined to the gene products of *trp E*, *trp D*, and *trp C*.
 - B The presence of lactose results in the expression of β -galactosidase.
 - C The very low concentrations of tryptophan in the cell results in the expression of β -galactosidase.
 - D Absence of *lac* promoter prevents the expression of β -galactosidase.
- 15 What are the correct characteristics for eukaryotic genome?

	Enclosed with a porous membrane	DNA compacted with proteins	Extrachromosomal DNA present	Repeat sequences uncommon
A	√	√	√	X
B	√	√	X	X
C	X	X	√	√
D	√	√	X	√

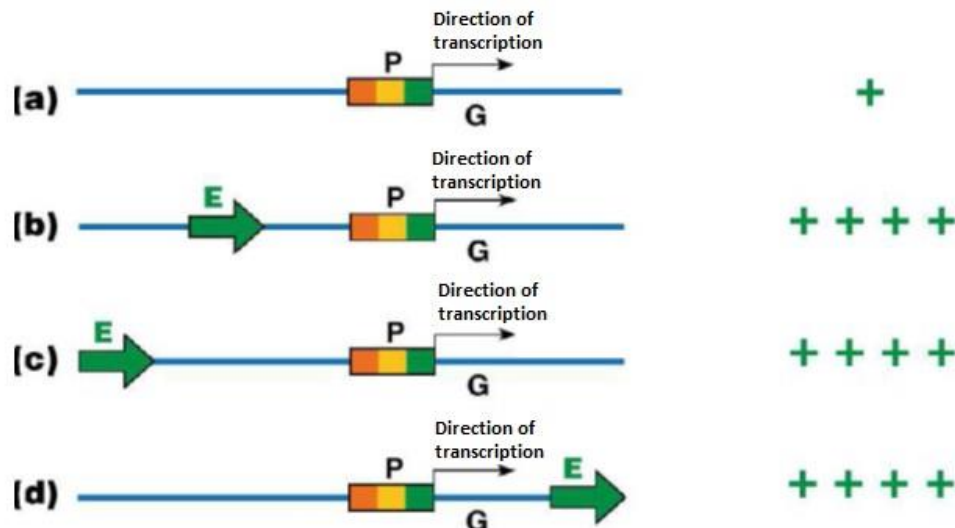
- 16 Which of the following observations support the development of cancer as a multistep process?

1. Cancer involves the accumulation of mutations in at least one proto-oncogene and several tumor suppressor genes in a specific order.
2. The lag time that often separates exposure to a cancer-causing agent and the development of cancer.
3. An increase risk of people who inherit predisposing mutations to developing cancer.
4. The incidence of cancer increases with age.

- A 2, 3 and 4
- B 1, 3 and 4
- C 2 and 4
- D 2 and 3

- 17** Recombinant DNA techniques can be used to alter the locations of control elements within DNA sequences, so as to study the effects of these changes on the levels of transcription. The diagram shows the various structures of transcription units and the corresponding relative levels of transcription.

The promoter, enhancer and coding sequences are represented by letters P, E and G respectively. The number of symbol '+' indicates the relative levels of transcription.



With reference to the diagram, which of these statements is a valid conclusion?

- A** The relative distance between promoter and enhancer has no effect on the rate of transcription.
- B** The rate of transcription is increased when the enhancer is located upstream of the promoter.
- C** An enhancer is required for transcription.
- D** A promoter bound with RNA polymerase and general transcription factors result in basal rate of transcription.
- 18** Which statement best explains how related genes involved in the same metabolic pathway are expressed together in eukaryotic cells?
- A** Related genes are usually located on the same chromosome so that they can be controlled by the same set of control elements.
- B** The same set of transcription factors may be capable of recognising the same promoter site of related genes.
- C** There are specific sets of control elements associated with related genes, recognised by specific sets of transcription factors.
- D** Within the control element of related genes, the specific numbers of transcription factors binding to the control element will enable related genes to be expressed.

- 19** In tomato plants, flower colour is controlled by a pair of alleles having different coding base sequences. One of the alleles codes for yellow flowers while another allele codes for white flowers. When pollen from a tomato plant with yellow flowers was transferred to a pure-bred plant with white flowers, offspring plants with yellow flowers and plants with white flowers were produced in approximately equal numbers.

Which of the following statements about the genetic cross described above is correct?

- A** The two alleles are located on the same chromosome.
 - B** In the parental plant with yellow flowers, both alleles are present but only the allele coding for yellow flowers is transcribed and translated as it is dominant.
 - C** The frequency of the allele coding for white flowers in the offspring plant population is 75%.
 - D** Repeated transfer of pollen from an offspring plant with yellow flowers to the parent plant with yellow flowers will never produce plants with white flowers.
- 20** Skin colour in a variety of pumpkins is controlled by two pairs of alleles, P/p and R/r , which segregate independently.

When dominant allele P is present, the pumpkin skin cells will produce pigments. Pumpkins without pigment are white. When dominant allele R is present, the pumpkin will produce red pigments. The recessive allele r results in pumpkins having yellow-coloured skin.

When a plant with yellow pumpkins is self-pollinated, a phenotypic ratio of 3 yellow : 1 white was obtained.

What would be the expected phenotypic ratio if the plant with yellow pumpkins was crossed with a plant pure breeding for red pumpkins?

- A** 1 red : 1 yellow
- B** 12 red : 3 yellow : 1 white
- C** all white
- D** all red

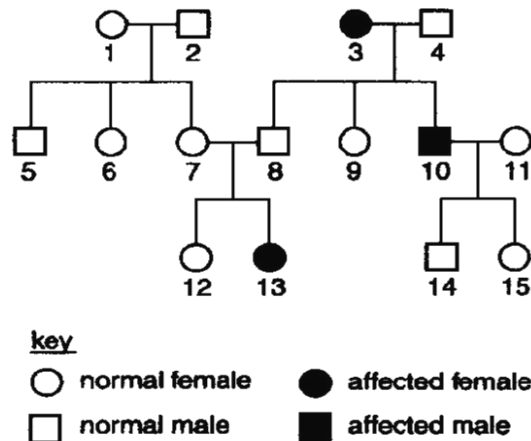
- 21 Two pure-bred lines of different plant varieties, which differed markedly in bean seed mass, were crossed. The mass of bean seeds produced by the two parental varieties and their offspring were measured to the nearest gram. The number of bean seeds in each mass category was counted.

The table below shows the results.

mass of bean / mg		51-150	151-250	251-350	351-450	451-550	551-650	651-750	751-850	851-950
number of beans	Parental	5	375	177				352	955	10
	Offspring			13	544	974	48	2		

Which statement **incorrectly** explains these experimental data?

- A The range of phenotypes for the characteristic of bean seed mass is continuous and is due to several different nucleotide sequences at different chromosomal positions controlling the characteristic.
- B The greater variation in bean seed mass observed in the offspring generation as compared to the parental generation is due to crossing over between homologous chromosomes, random fusion of gametes, different survival rates of the gametes and zygotes.
- C The phenotypic effects of different nucleotide sequences at different chromosomal positions can be summated to determine the bean seed mass in each plant.
- D Various environmental factors affect the mass of bean seeds in plants.
- 22 The family tree shows the inheritance of a condition caused by the change in nucleotide sequence of gene R.



What is the probability that the first child is an affected girl when individual 7 mated with an affected male instead?

- A 0
- B 0.25
- C 0.50
- D 0.75

23 A naturally occurring mutant tomato plant with yellow fruit was crossed with the red wild type.

The first generation plants were self-pollinated and were also backcrossed with both parents. The results are below.

- The first generation plants all produced red coloured fruits.
- The second generation plants produced fruits in a ratio of 3 red : 1 yellow.
- Plants from a backcross with red wild type all produced red fruits.
- Plants from a backcross with the yellow mutant produced fruits in a ratio of 1 red : 1 yellow.
- The mutant yellow fruit had a higher rate of chlorophyll breakdown during fruit ripening.
- The red fruit from the first generation showed lower rates of chlorophyll breakdown, as similarly observed in red wild type fruits.

From the results above, what can be concluded?

- I The dominant allele coding for red fruits is epistatic to the recessive allele coding for yellow fruits.
- II For plants produced from the backcross with red wild type, there is equal proportion of plants which are homozygous and plants which are heterozygous at the gene locus controlling fruit colour.
- III The dominant allele coding for red fruits influences chlorophyll metabolism in fruits.
- IV The effect of the allele coding for red fruits on chlorophyll metabolism is inhibited by the allele coding for yellow fruits.

- A II and III
- B III and IV
- C I, II and IV
- D I, III and IV

- 24 Two genes involved in coat colour of goats are at loci on different chromosomes.

The colour gene C causes the hairs to have uniform colour and has three alleles.

- C^{DB} giving dark brown hairs
- C^B giving black hairs
- C^{MB} giving medium brown hairs

A dominant allele of the agouti gene (A^G) causes the development of white hairs between the coloured hairs giving the coat a shaded appearance.

The table shows the results of crosses between a male goat and two female goats.

parents	offspring
black agouti male x uniformly dark brown female	50% uniform, 50% agouti; 50% dark brown, 50% black
black agouti male x medium brown agouti female	all agouti; 50% black, 50% medium brown

Which row shows the possible genotypes of the male and female goats?

	black agouti male	uniformly dark brown female	medium brown agouti female
A	$C^B C^{MB} A^G A^G$	$C^B C^{DB} A^G A^g$	$C^{MB} C^{MB} A^G A^G$
B	$C^B C^{MB} A^G A^g$	$C^B C^{DB} A^g A^g$	$C^B C^{MB} A^G A^G$
C	$C^B C^{DB} A^G A^G$	$C^{DB} C^{MB} A^G A^g$	$C^B C^{DB} A^G A^g$
D	$C^B C^{MB} A^G A^g$	$C^B C^{DB} A^g A^g$	$C^{MB} C^{MB} A^G A^G$

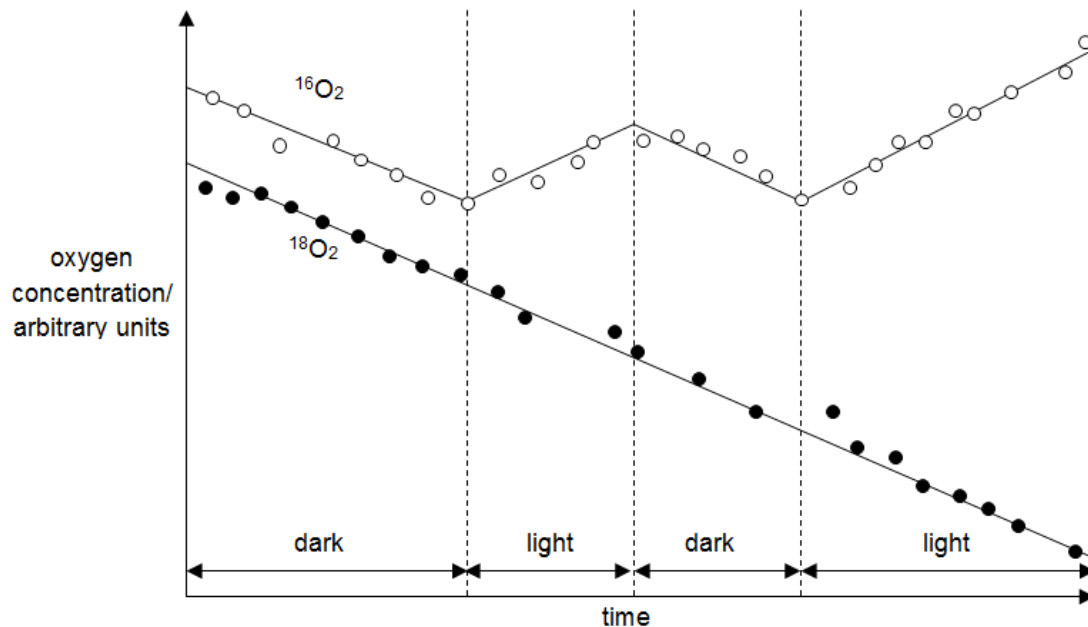
- 25 Which of the following statement(s) is/are true regarding cyclic and non-cyclic photophosphorylation?

- 1 Cyclic photophosphorylation requires light at 680nm to function optimally while non-cyclic photophosphorylation requires light at both 680nm and 700nm to function optimally.
- 2 Only non-cyclic photophosphorylation will be affected in the absence of NADP reductase.
- 3 Only non-cyclic photophosphorylation utilises electrochemical gradient generated to drive ATP production.
- 4 Only cyclic photophosphorylation can function in the absence of photosystem II.

- A 4 only
 B 2 and 4 only
 C 2 and 3 only
 D 1, 2 and 4 only

- 26 Isotopes of oxygen can be used to distinguish between oxygen absorbed by plants and oxygen evolved.

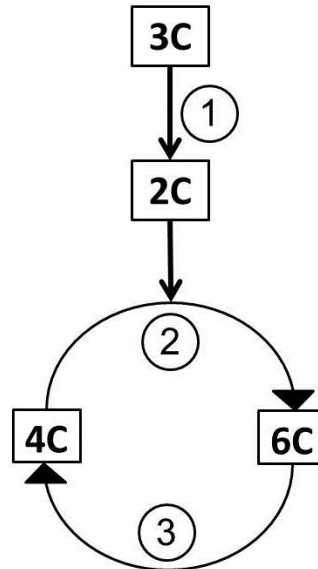
A mixture of oxygen isotopes $^{16}\text{O}_2$ and $^{18}\text{O}_2$ was supplied to a suspension of the unicellular alga *Chlorella* which had previously been exposed to $^{16}\text{O}_2$ only. During the following hour, changes in concentration of these gases in the suspension were measured in light and dark conditions. The graph shows the results.



What caused the concentration of $^{16}\text{O}_2$ to rise in light?

- A H_2^{18}O was being photolysed more rapidly than H_2^{16}O .
- B $^{16}\text{O}_2$ was being produced in photosynthesis faster than it was being absorbed in respiration.
- C $^{16}\text{O}_2$ was being absorbed at different rates in light and dark.
- D $^{16}\text{O}_2$ was being produced in photosynthesis but was not being absorbed in respiration.

- 27 The diagram shows some phases of respiration and numbers of carbon atoms in the compounds involved.



Which reactions occur at Phase 1-3?

	Phase 1	Phase 2	Phase 3
A	Oxidative decarboxylation and oxidation of coenzyme	Substrate-level phosphorylation	CO ₂ production and oxidative phosphorylation
B	CO ₂ production and reduction of coenzyme	ATP hydrolysis	Oxidative decarboxylation and dehydrogenation
C	Oxidative decarboxylation and reduction of coenzyme	Substrate-level phosphorylation	CO ₂ production and dehydrogenation
D	CO ₂ production and oxidation of coenzyme	ATP hydrolysis	Oxidative decarboxylation and oxidative phosphorylation

28 Two test tubes containing the following contents are shown below:

Tube 1:

Radioactive glucose solution + yeast cells suspension + oxygen + antimycin

Tube 2:

Radioactive glucose solution + yeast cells suspension + oxygen

Radioactive glucose has all its six carbons made of radioactive ^{14}C . The initial radioactivity measured for the glucose in each test tube is 60 arbitrary units.

Antimycin is an electron transport chain inhibitor.

If the gaseous product and the aqueous products are tested using a radioactive meter after all the glucose has been metabolised, what would be the final observed readings?

	Tube 1 (radioactivity measured/ arbitrary units)		Tube 2 (radioactivity measured/ arbitrary units)	
	aqueous products	gaseous products	aqueous products	gaseous products
A	0	60	40	20
B	20	40	0	60
C	40	20	0	60
D	40	20	60	0

29 Some statements about the action potential generation in a neurone cell and nerve impulse transmission are listed below.

- I** Sodium-potassium pump increases membrane potential during hyperpolarisation to restore membrane potential to -70 mV.
- II** Depolarisation of the membrane only involves increase in cytoplasmic concentration of Na^+ ions.
- III** When the membrane is depolarised, the outside of membrane is +40 mV compared to the inside.
- IV** Restricting Na^+ and K^+ ion diffusion to certain regions of the membrane increases the number of nerve impulses per unit time.

Which of the statements above are true?

- A** I and III
- B** II and IV
- C** I, II and IV
- D** All of the above

- 30 Four proteins isolated from a human cell were investigated for their involvement in cell signalling pathways.

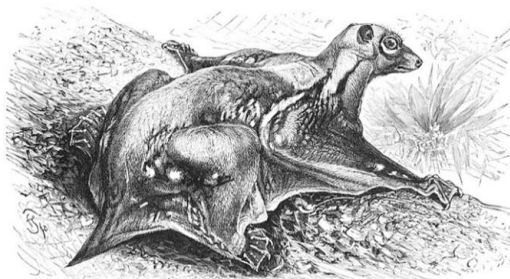
	Protein A	Protein B	Protein C	Protein D
Transmembrane domain	-	+	+	-
DNA binding domain	-	-	-	+
Enzymatic domain	+	-	+	-

Key: (+) = present , (-) = absent

Which of the following shows the correct identity of these four proteins?

	Protein A	Protein B	Protein C	Protein D
A	GPCR	Ras protein	RTK	Testosterone receptor
B	Ras protein	RTK	GPCR	Testosterone receptor
C	Testosterone receptor	GPCR	RTK	Ras protein
D	Ras protein	GPCR	RTK	Testosterone receptor

- 31 The Malayan Colugo is a forest dwelling creature that relies on closely spaced trees to glide from one area to another in the primary forest. They do this by a thin membrane that stretches from the side of the neck to the tips of the fingers and toes, and down to the tip of the tail.

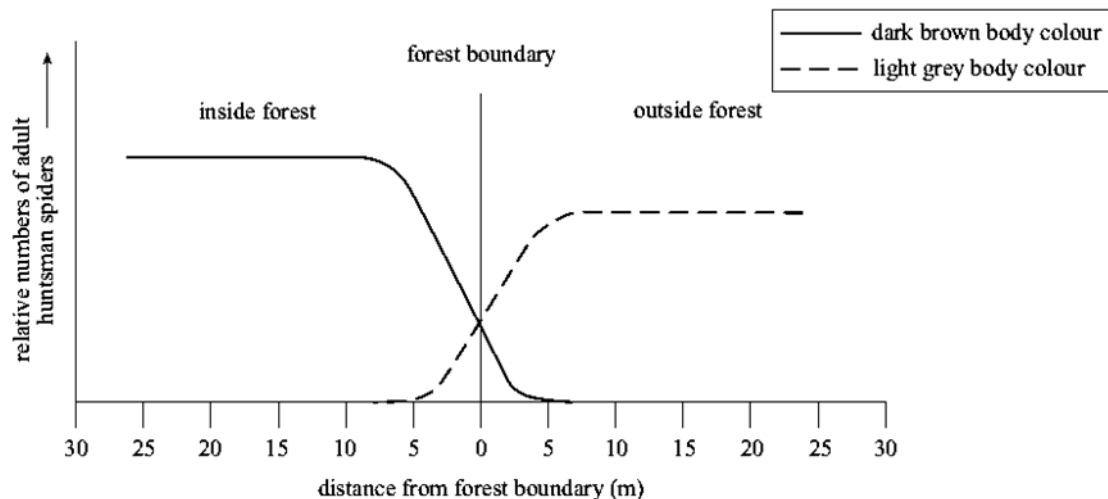


The population of the Malayan Colugo in Singapore are descendants from a small population of Colugos that were over-hunted and had their habitats destroyed post-1819 when 90% of primary forests in Singapore were cleared leaving only pockets of rainforests fragments separated from each other by urban developments.

Which of the following best describes the evolution of the Malayan Colugo in Singapore over two centuries?

	Genetic variability	Evolutionary process	Type of speciation
A	low variability	population bottleneck	allopatric
B	low variability	founder effect	sympatric
C	high variability	disruptive selection	allopatric
D	high variability	population bottleneck	sympatric

- 32 The graph below shows the distribution of huntsman spiders at a forest boundary:



One species of huntsman spider (*Isopeda isopedella*) varies in body colour from dark brown to light grey. In one community at the forest boundary, two populations of this species were found. Some were found living inside the forest and others were found living just outside the forest. The relative numbers of dark brown adult spiders and light grey adult spiders found at certain distances from the forest boundary are shown in the graph above.

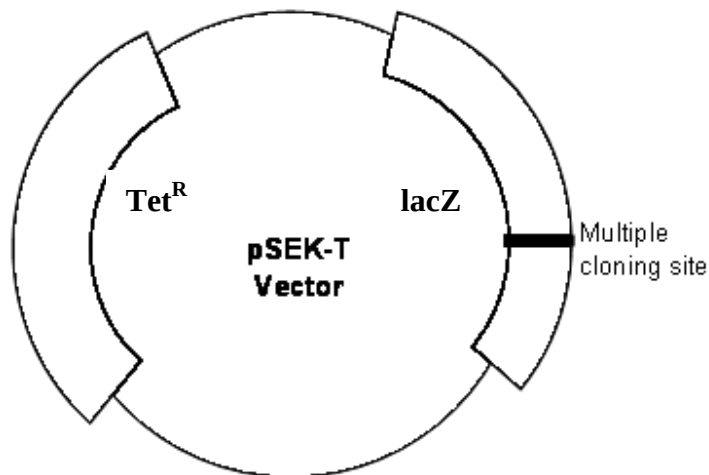
Which of the following can be **least** inferred from the graph?

- A At the forest boundary, the selection pressure is stronger than that either inside the forest or outside the forest.
 - B The plateau in the population number as seen inside the forest and outside the forest is due to competition among the adult spiders for resources.
 - C The lower plateau of spider population outside the forest compared to that of inside the forest is due to the presence of additional selection pressure existing outside the forest.
 - D Dark brown huntsman spiders are not eaten by birds inside the forest as their colour allows them to camouflage and hence provides a selective advantage.
- 33 Giant tortoises had been previously classified based on morphological differences, primarily using carapace (shell) shape, which varies from domed to saddleback with intermediate forms also occurring.

Which of the following statement(s) explain why classification using carapace shape may not be an accurate way of determining the evolutionary relationships?

- A It is based only on a single trait.
- B Carapace shape may be analogous structures due to convergent evolution of two different species.
- C It is subjective as there is continuous variation of carapace shape with no distinct phenotypes.
- D All of the above.

- 34 The diagram below shows a pSEK-T vector and the position of its multiple cloning site and selectable markers.



If the gene of interest were to be inserted into the multiple cloning site, what should be added to the agar plate in order to screen for transformed bacteria colonies containing recombinant plasmids and how would the colonies appear?

	Chemicals to be added to agar		Colour of colonies
A	tetracycline	X-gal	blue
B	β -galactosidase	X-gal	blue
C	tetracycline	X-gal	white
D	β -galactosidase	lactose	white

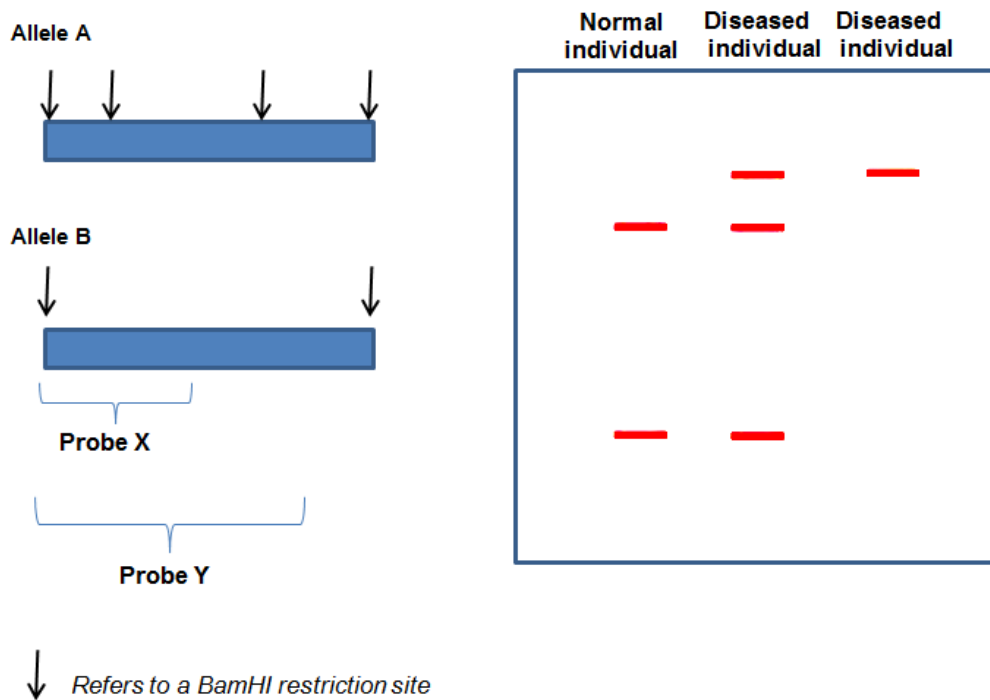
- 35 In humans, the CFTR gene codes for a transmembrane protein needed for the diffusion of Cl^- ions into and out of epithelial cells. More than 1000 types of mutation of the CFTR gene have been discovered.

Which of the following statements are correct?

- 1 CFTR gene is polymorphic.
- 2 The gene locus for CFTR is multiple allelic.
- 3 The gene locus for CFTR is polygenic.
- 4 Allele frequency of the CFTR gene in human population is more than 1000.

- A** 1 and 2
B 2 and 3
C 1 and 4
D 3 and 4

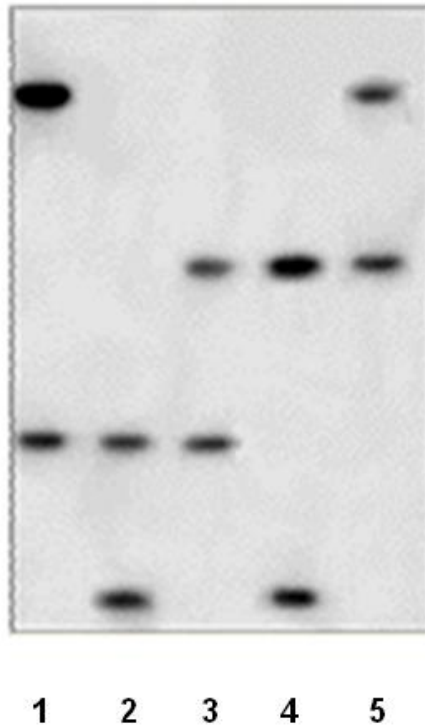
- 36 The DNA of 3 individuals in a family affected by a disease caused by a mutation in an autosomal gene was analysed using restriction fragment length polymorphism (RFLP). The figures below shows the restriction sites of two alleles suspected to cause the disease and the resulting autoradiograph obtained upon RFLP analysis.



Which of the following combinations is correct?

	Allele causing the disease	Nature of the mutation	Probe used in the analysis
A	Allele A	Dominant	Probe X
B	Allele A	Recessive	Probe Y
C	Allele B	Dominant	Probe X
D	Allele B	Recessive	Probe Y

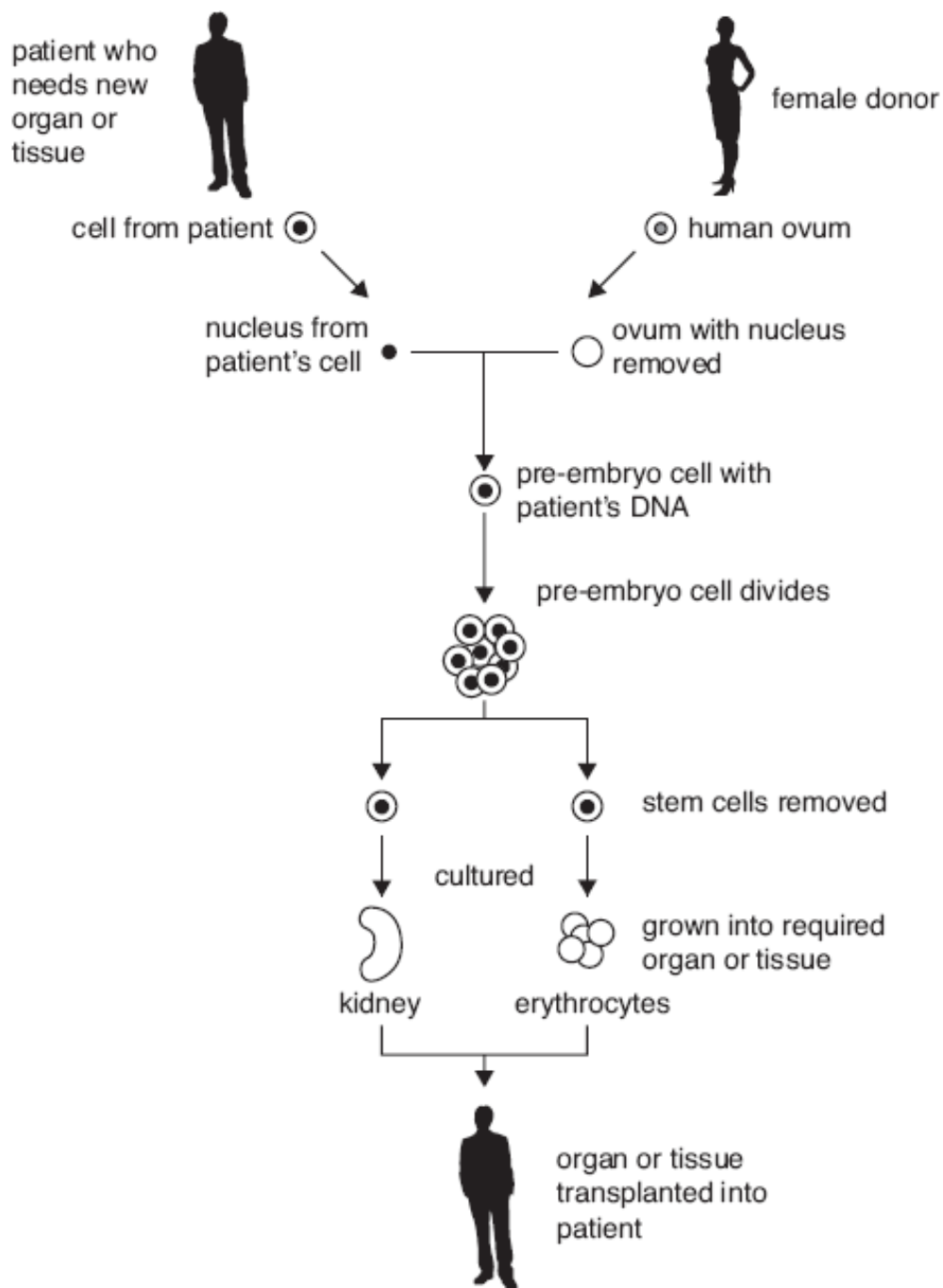
- 37 Results from the analysis of a single variable number tandem repeat (VNTR) locus for a man and four of his children are shown in the autoradiograph.



Which of the following is the **most** likely conclusion from these results?

- I Lane 3 most probably contains the DNA of the father.
 - II The father and children relationship cannot be determined from these results since only one VNTR locus is analysed.
 - III This analysis is based on differences in the number and length of restriction fragments generated in different individuals.
- A I only
B II only
C I and III only
D II and III only

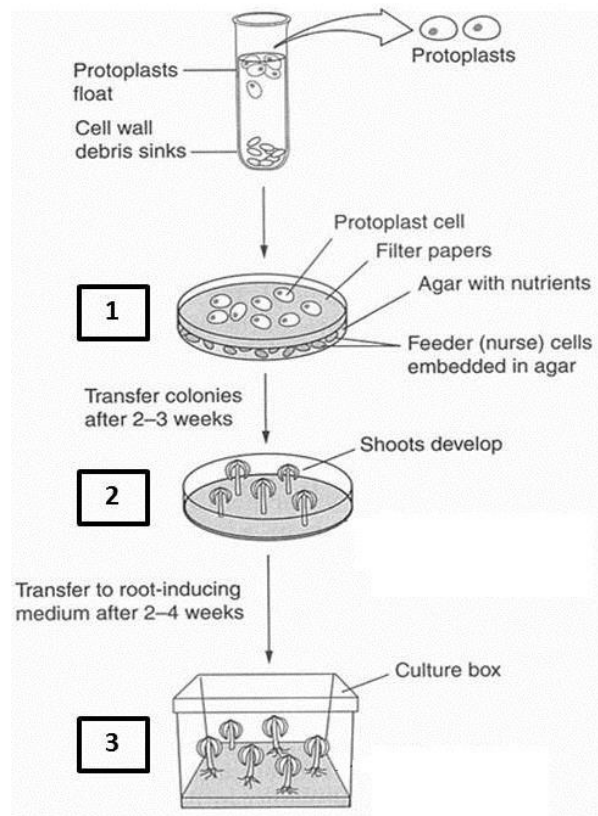
- 38** In the future, therapeutic cloning may produce new tissues and organs to treat diseases. The figure below shows stages in producing new tissues and organs by therapeutic cloning.



Which of the following statement about therapeutic cloning is correct?

- A** The above procedure is a form of germ-line gene therapy.
- B** The ovum containing the patient's DNA is a diploid cell.
- C** The pre-embryo cells are multipotent cells because they contain an adult patient's DNA.
- D** Therapeutic cloning in this manner is more ethical than using stem cells directly from a donor.

- 39 The diagram below shows steps of the procedure that can be used to produce plantlets from protoplast cells isolated from leaves.



Which of the following auxin:cytokinin ratios are correctly employed in growth medium for the respective stages of plant tissue culture?

	1	2	3
A	2 : 0.5	2 : 0.2	2 : 0.02
B	2 : 0.2	2 : 0.02	2 : 0.5
C	2 : 0.02	2 : 0.2	2 : 0.5
D	2 : 0.2	2 : 0.5	2 : 0.02

- 40 Some scientists are concerned about the release of genetically modified microorganisms into their natural habitats.

What is the **most** likely reason for this concern?

- A The microorganisms may reproduce quickly.
- B The microorganism may not survive.
- C The mutation rate of the microorganism would increase.
- D The transfer of changed genes to other organism.

Paper 1 Answer scheme

Qns	Ans
1	B
2	D
3	C
4	D
5	C
6	C
7	D
8	C
9	C
10	B
11	C
12	C
13	B
14	C
15	A
16	A
17	A
18	C
19	C
20	D

Qns	Ans
21	B
22	B
23	A
24	D
25	B
26	B
27	C
28	C
29	C
30	D
31	A
32	D
33	D
34	C
35	A
36	C
37	A
38	B
39	D
40	D